

Adaptive Estimation of the Pennes' Bio-Heat Equation - II: a NN-based implementation for real-time applications

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*Companion Paper: Adaptive Estimation of the Pennes' Bio-Heat Equation - I:
Observer Design [WeB26.1]*

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Abstract

- Motivation: improving accuracy and outcomes of robot-assisted oncological hyperthermia treatments;
- Concept: observing heat transfer in biological tissues using boundary measurements and controls to achieve real-time temperature estimation in a 1D domain;
- Approach: employing Physics-Informed Neural Networks combined with state observer to estimate uncertain parabolic reaction-diffusion Partial Differential Equations (PDEs).

Superficial hyperthermia in a nutshell

- heating tumours in the 41-43 °C range is a powerful radio- and chemo-sensitizer;
- heat is delivered by a microwave radiating antenna;
- water bolus between the applicator and the patient is used to prevent skin burns;
- superficial temperature measurements are easily obtained through a matrix of thermocouples.



Superficial Hyperthermia: does it work?

Literature reports 1.5-factor improvement in complete response, local control and survival

Clinical treatment success hinges upon:

- achieving and maintaining high target temperatures at appropriate timings and durations: each 0.5°C higher effectively doubles the thermal dose;
- ensuring reproducibility across successive treatments.

The main difficulties are:

- unknown and variable blood flow entering the heated region causes cold tracks;
- target temperatures only known through invasive measurement and at specific points.

Objective: feedback control of treatments through non-invasive temperature estimation in the presence of uncertain coefficients.

Bio-Heat Equation

Pennes' Bio-Heat Equation is scaled using typical treatment parameter values:

$$a_1 \partial_\tau \theta = \partial_{XX} \theta - a_2 W \theta + a_3 Q \quad (1)$$

$$\theta(0, \tau) = C_0 \quad \tau \geq 0$$

$$\partial_X \theta(1, \tau) = v(\tau) \quad \tau \geq 0 \quad (2)$$

$$\theta(X, 0) = \theta_0(X) \quad X \in [0, 1]$$

- $\theta = \frac{T - T_a}{T_M - T_a}$: scaled temperature;
- $X = \frac{x}{L_0}$, $\tau = \frac{t}{\tau_f}$: normalized spatial and temporal coordinates;
- a_1 , a_2 , a_3 : constant parameters depending on tissue properties;
- W : perfusion rate, unknown but variable in a known range for a given tissue type;
- Q : sum of metabolic heat and power deposited by the applicator.

Observer Design

Under the assumption that:

- the input $v(\tau) = \partial_X \theta(1, \tau)$ is accessible and the output $y(\tau) = \theta(1, \tau)$ is available at any $\tau \geq 0$;
- W is constant and $0 \leq W_{\min} \leq W \leq W_{\max}$.

With some guess $\tilde{W} \in [W_{\min}, W_{\max}]$, the observer can be designed as follows:

$$a_1 \partial_\tau \hat{\theta} = \partial_{XX} \hat{\theta} - a_2 \tilde{W} \hat{\theta} + a_3 \mathcal{Q} \quad (3)$$

$$\hat{\theta}(0, \tau) = C_0 \quad \tau \geq 0$$

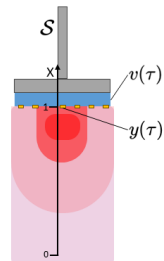
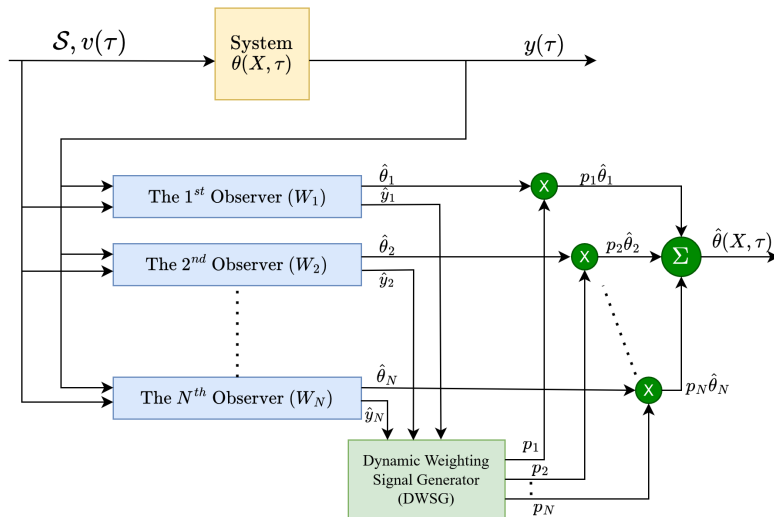
$$\partial_X \hat{\theta}(1, \tau) = v(\tau) + \alpha(y(\tau) - \hat{\theta}(1, \tau)) \quad \tau \geq 0 \quad (4)$$

$$\hat{\theta}(X, 0) = \hat{\theta}_0(X) \quad X \in [0, 1]$$

Being $\alpha > 0$ the output injection gain, and $\hat{\theta}_0$ subject to compatibility conditions.

The error $e_\theta(X, \tau) = \theta(X, \tau) - \hat{\theta}(X, \tau)$ is ultimately bounded and the convergence towards the attractive set is exponential.

Multiple-model Observer [1]



Overall estimator

The overall estimator is a weighted average of each observer's prediction, with weights $p_i(\tau)$ being updated according to

$$\dot{p}_i(\tau) = -\lambda \left(1 - \frac{e^{-\mu_i(\tau)}}{\sum_{\ell=1}^N p_\ell(\tau) e^{-\mu_\ell(\tau)}} \right) p_i(\tau) \quad i = 1, \dots, N \quad (5)$$

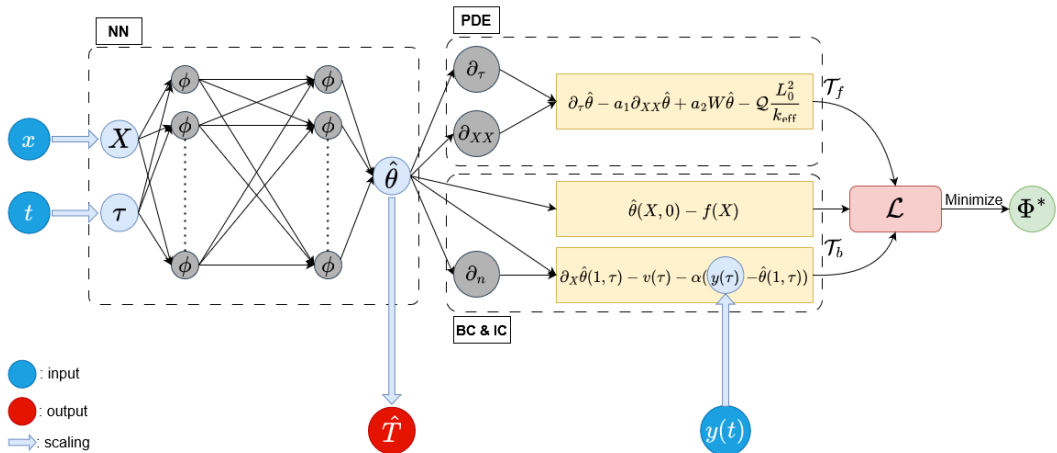
where

$$\mu_i(\tau) := \xi |y(\tau) - \hat{y}_i(\tau)|^2 \quad \xi > 0 \quad (6)$$

is the absolute output error and $\lambda > 0$ is the adaptive gain.

Neural Bio-Heat Observer (NBHO)

Each of the N observers in the Multiple-model has been implemented using Physics-Informed Neural Networks (PINNs) utilizing the DeepXDE [2] Python library



Advantages in using PINNs

Computational issues arising in the integration of multiple PDEs and ODE are handled by reducing the computation of the absolute output error $\mu_i(\tau)$ to a forward pass of a deep neural network (few milliseconds).

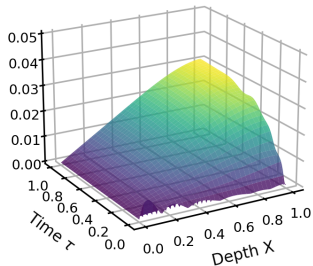
Moreover, they present some important features for further improvements:

- can potentially scale to multiple input dimensions seamlessly. Alternatively, the use of a solver for each setting is necessary;
- no spatiotemporal grid: the system is probed continuously.

Challenges in using PINNs

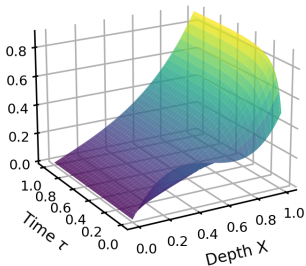
Accuracy of PINNs solution depends on hyperparameters.

Error after HPO
RMSE: 0.0165
95th Percentile Error: 0.0290

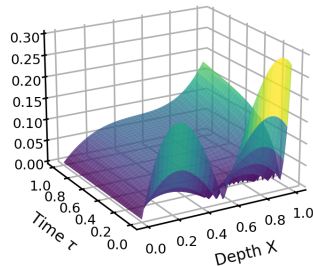


[1e-03, 4, 50, "sin", "Glorot normal", 100, 1, 1]

Ground Truth



Error before HPO
RMSE: 0.0703
95th Percentile Error: 0.1323



[2e-03, 1, 5, "relu", "He normal", 10, 250, 227]

Hyperparameter Optimization (HPO) using `skopt gp.minimize`

Numerical simulation

- $W = 2.3 \frac{kg}{m^3 s}$;
- Homogeneous medium (muscle tissue);
- 8 observers with $\tilde{W}_j \in [0.45, 4] \frac{kg}{m^3 s}$;
- $Q = 0$, boundary conditions to simulate heat source;
- $\alpha = 4$, $\xi = 1$, $\lambda = 200$.

Networks structure: 4-layer perceptrons, 20 hidden neurons per layer; *tanh* activation function and parameters initialization according to the *Glorot uniform* method; training for 20000 iterations with SGD Adam solver, learning rate = 1e-03.

Goal: Reconstruct asymptotically $\theta(X, \tau)$, given the measured $y(\tau) = \theta(1, \tau)$ and knowing the flux $v(\tau)$ in the same boundary.

Note: the adaptation rate shown in the following simulation result was impossible to achieve using Matlab pdepe function.

Results

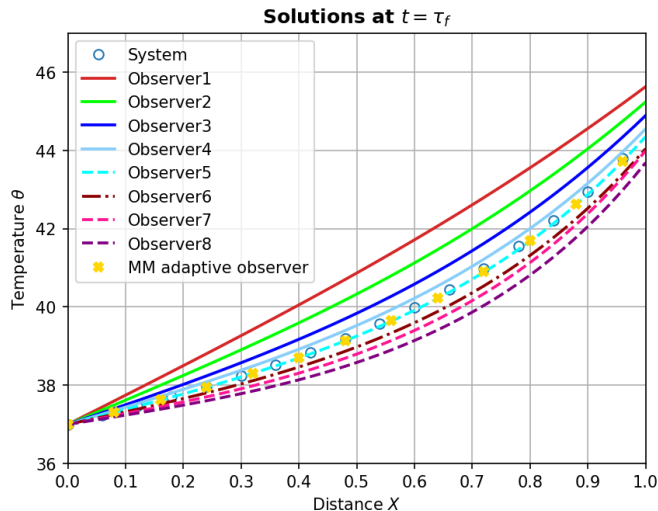


Figure: Solution vs. observers at $\tau = t_f$

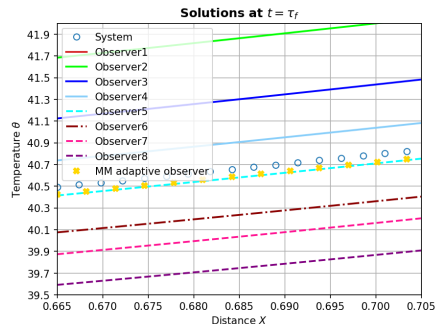


Figure: Zoomed version

Results (2)

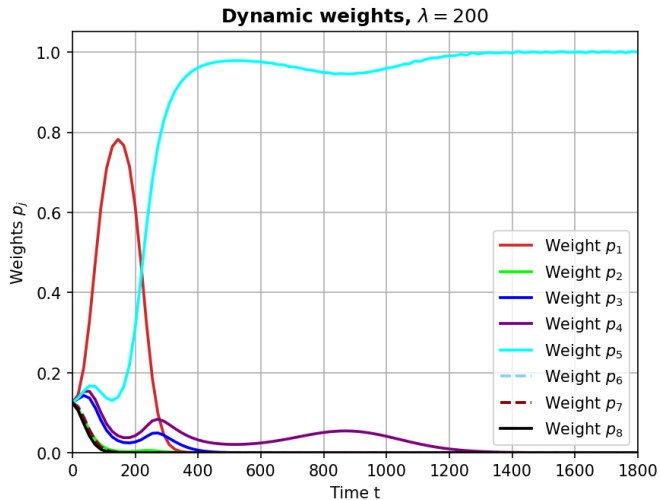


Figure: Dynamic weights

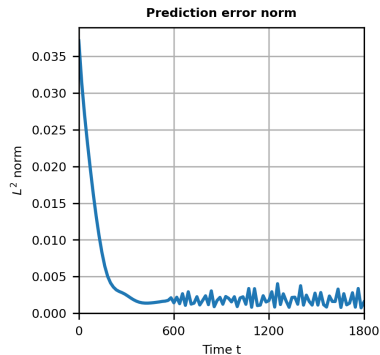


Figure: Observation error

Conclusion and Further Development

- Multiple-model observer to reconstruct the entire system profile and retrieve unknown reaction coefficient based on boundary measurements;
- Limitations of classical numerical simulation tools are handled through an implementation of the estimator relying on Physics-Informed Neural Networks.

Work in progress:

- Introduction of the source term in the PBHE;
- Augmentation of PINNs' dimensionality to consider: superficial flux as a control, and endpoint temperature measurement;
- Experimental validation;

Bibliography

- [1] Vahid Hassani, A Pedro Aguiar, António M Pascoal, and Michael Athans. Further results on plant parameter identification using continuous-time multiple-model adaptive estimators. In *Proceedings of the 48h IEEE Conference on Decision and Control held jointly with the 2009 28th Chinese Control Conference*, Shanghai, China, December 15-18, 2009.
- [2] M. Raissi, P. Perdikaris, and G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.